

GOOGLE CLOUD PROFESSIONAL CLOUD ARCHITECT

Section 3.5 — Deploying and Implementing Networking Resources

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Study Notes Study Notes April 2026

Exam Relevance

This topic is part of **Section 3: Deploying and implementing a cloud solution (~25 % of the exam)**. You must know how to create VPCs with subnets, configure firewall rules and policies, and set up VPN and VPC peering.

1. Creating a VPC with Subnets

VPC Fundamentals

- A VPC is a **global resource** that spans all GCP regions
- Subnets are **regional resources** (span all zones within a region)
- Every GCP project has a **default VPC** with auto-mode subnets

Auto Mode vs. Custom Mode VPC

Feature	Auto Mode	Custom Mode
Subnet creation	Automatic (one per region)	Manual
IP ranges	Predefined (10.128.0.0/9)	You choose
New region subnets	Added automatically	Must add manually
Production use	Not recommended	Recommended
Control	Limited	Full

Creating VPCs

```
# Create a custom mode VPC
gcloud compute networks create my-vpc \
  --subnet-mode=custom \
  --bgp-routing-mode=regional

# Create an auto mode VPC
gcloud compute networks create my-auto-vpc \
  --subnet-mode=auto

# Convert auto mode to custom mode (one-way, irreversible)
gcloud compute networks update my-auto-vpc \
  --switch-to-custom-subnet-mode

# Delete a VPC
gcloud compute networks delete my-vpc
```

Creating Subnets

```
# Create a subnet
gcloud compute networks subnets create my-subnet \
  --network=my-vpc \
  --region=us-central1 \
  --range=10.0.1.0/24

# Create a subnet with secondary ranges (for GKE pods/services)
gcloud compute networks subnets create gke-subnet \
  --network=my-vpc \
  --region=us-central1 \
  --range=10.0.0.0/24 \
  --secondary-range=pods=10.4.0.0/14,services=10.8.0.0/20

# Create a subnet with Private Google Access
gcloud compute networks subnets create private-subnet \
  --network=my-vpc \
  --region=us-central1 \
  --range=10.0.2.0/24 \
  --enable-private-ip-google-access

# Create a subnet with flow logs
gcloud compute networks subnets create logged-subnet \
  --network=my-vpc \
  --region=us-central1 \
  --range=10.0.3.0/24 \
  --enable-flow-logs

# List subnets
gcloud compute networks subnets list --network=my-vpc

# Describe a subnet
gcloud compute networks subnets describe my-subnet --region=us-central1
```

Private Google Access

- Allows VMs with **only internal IPs** to access Google APIs and services
- Required for VMs without external IPs to reach Cloud Storage, BigQuery, etc.
- Enabled at the **subnet level**

```
# Enable Private Google Access on a subnet
gcloud compute networks subnets update my-subnet \
  --region=us-central1 \
  --enable-private-ip-google-access
```

VPC Flow Logs

- Capture network flow data for monitoring and forensics
- Logs IP, port, protocol, bytes, packets for each flow
- Can export to Cloud Logging, BigQuery, or Cloud Storage

2. Shared VPC

What Is Shared VPC?

Shared VPC allows an **organization** to share a VPC network across multiple projects.

```
Organization
├── Host Project (owns the Shared VPC)
│   ├── Shared VPC Network
│   │   ├── Subnet A (us-central1)
│   │   └── Subnet B (europe-west1)
│   ├── Service Project 1 (uses subnets from Shared VPC)
│   ├── Service Project 2 (uses subnets from Shared VPC)
│   └── Service Project 3 (uses subnets from Shared VPC)
```

Key Concepts

- **Host Project** — Contains the Shared VPC network and subnets
- **Service Projects** — Projects that use subnets from the host project
- Network administration is centralized in the host project
- Service projects can create resources (VMs, GKE, etc.) in shared subnets

Setting Up Shared VPC

```
# Enable Shared VPC on the host project
gcloud compute shared-vpc enable HOST_PROJECT_ID

# Associate a service project
gcloud compute shared-vpc associated-projects add SERVICE_PROJECT_ID \
  --host-project=HOST_PROJECT_ID

# Grant a user permission to use shared subnets
gcloud projects add-iam-policy-binding HOST_PROJECT_ID \
  --member="user:developer@example.com" \
  --role="roles/compute.networkUser"

# List service projects
gcloud compute shared-vpc list-associated-resources HOST_PROJECT_ID
```

When to Use Shared VPC

- Centralized network management
 - Consistent firewall rules across projects
 - Different teams/projects needing the same network
 - Billing separation with shared networking
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3. Firewall Rules and Policies

Firewall Rule Basics

- Applied at the **VPC level**
- **Stateful** — If ingress is allowed, return traffic is automatically allowed
- Default rules: Allow egress to all, deny ingress from all (except from within the VPC)
- Rules are evaluated by **priority** (0 = highest priority, 65535 = lowest)

Creating Firewall Rules

```
# Allow HTTP traffic from any source
gcloud compute firewall-rules create allow-http \
  --network=my-vpc \
  --allow=tcp:80 \
  --source-ranges=0.0.0.0/0 \
  --target-tags=http-server \
  --priority=1000 \
  --direction=INGRESS

# Allow HTTPS traffic
gcloud compute firewall-rules create allow-https \
  --network=my-vpc \
  --allow=tcp:443 \
  --source-ranges=0.0.0.0/0 \
  --target-tags=https-server \
  --priority=1000 \
  --direction=INGRESS

# Allow SSH from a specific IP range
gcloud compute firewall-rules create allow-ssh-office \
  --network=my-vpc \
  --allow=tcp:22 \
  --source-ranges=203.0.113.0/24 \
  --priority=1000 \
  --direction=INGRESS

# Allow internal traffic between subnets
gcloud compute firewall-rules create allow-internal \
  --network=my-vpc \
  --allow=tcp,udp,icmp \
  --source-ranges=10.0.0.0/8 \
  --priority=1000 \
  --direction=INGRESS

# Allow IAP tunneling for SSH
gcloud compute firewall-rules create allow-iap-ssh \
  --network=my-vpc \
  --allow=tcp:22 \
  --source-ranges=35.235.240.0/20 \
  --priority=1000 \
  --direction=INGRESS

# Deny all egress to a specific destination
gcloud compute firewall-rules create deny-external \
  --network=my-vpc \
  --action=DENY \
  --rules=all \
  --destination-ranges=0.0.0.0/0 \
```

```
--priority=1000 \  
--direction=EGRESS  
  
# List firewall rules  
gcloud compute firewall-rules list --filter="network=my-vpc"  
  
# Describe a firewall rule  
gcloud compute firewall-rules describe allow-http  
  
# Delete a firewall rule  
gcloud compute firewall-rules delete allow-http
```

Firewall Rule Components

Component	Description	Example
Direction	INGRESS (incoming) or EGRESS (outgoing)	--direction=INGRESS
Priority	0-65535 (lower = higher priority)	--priority=1000
Action	ALLOW or DENY	--allow=tcp:80 or --action=DENY
Targets	Which instances the rule applies to	Tags, service accounts, or all
Source (ingress)	Where traffic comes from	IP ranges, tags, SAs
Destination (egress)	Where traffic goes to	IP ranges
Protocol/Port	Which traffic	tcp:80 , udp:53 , icmp

Target Specification Methods

1. Network Tags

```
# Rule targets instances with the tag "web-server"
gcloud compute firewall-rules create allow-web \
  --network=my-vpc \
  --allow=tcp:80,tcp:443 \
  --target-tags=web-server

# Apply tag to an instance
gcloud compute instances create my-vm \
  --tags=web-server \
  --zone=us-central1-a
```

2. Service Accounts

```
# Rule targets instances running with a specific service account
gcloud compute firewall-rules create allow-db-access \
  --network=my-vpc \
  --allow=tcp:3306 \
  --target-service-accounts=db-server@PROJECT_ID.iam.gserviceaccount.com \
  --source-service-accounts=web-server@PROJECT_ID.iam.gserviceaccount.com
```

3. All Instances (No Target Specified)

If no target is specified, the rule applies to **all instances** in the VPC.

Network Tags vs. Service Accounts for Firewall Rules

Feature	Network Tags	Service Accounts
Management	Applied per instance	IAM-managed
Security	Any instance admin can change tags	Requires IAM permission to assign
Granularity	Instance-level	Service identity level
Recommended	Simple setups	Production (more secure)

Hierarchical Firewall Policies

- Applied at the **organization or folder level**
- Inherited downward through the hierarchy
- Evaluated **before** VPC firewall rules

- Useful for organization-wide security policies

4. VPC Network Peering

What Is VPC Peering?

- Connects two VPC networks so resources can communicate using **internal IPs**
- Works across projects and organizations
- Traffic stays on Google's network (never traverses the public internet)

Key Properties

- **Non-transitive** — If VPC A peers with VPC B, and VPC B peers with VPC C, A and C cannot communicate through B
- **Symmetric** — Both sides must configure peering
- **No overlapping IP ranges** — Peered VPCs cannot have overlapping CIDR ranges
- No network bandwidth restrictions (uses Google's backbone)
- Each VPC can have up to **25 peering connections** (configurable)

Setting Up VPC Peering

```
# From VPC-A, peer with VPC-B
gcloud compute networks peerings create peer-a-to-b \
  --network=vpc-a \
  --peer-network=vpc-b \
  --peer-project=PROJECT_B_ID

# From VPC-B, peer with VPC-A (both sides required)
gcloud compute networks peerings create peer-b-to-a \
  --network=vpc-b \
  --peer-network=vpc-a \
  --peer-project=PROJECT_A_ID

# List peering connections
gcloud compute networks peerings list --network=vpc-a

# Delete a peering connection
gcloud compute networks peerings delete peer-a-to-b --network=vpc-a
```

Export/Import Custom Routes

```
# Export custom routes to peered network
gcloud compute networks peerings update peer-a-to-b \
  --network=vpc-a \
  --export-custom-routes

# Import custom routes from peered network
gcloud compute networks peerings update peer-a-to-b \
  --network=vpc-a \
  --import-custom-routes
```

5. Cloud VPN

What Is Cloud VPN?

- Connects your on-premises network to GCP VPC over an **encrypted IPsec tunnel** through the public internet

VPN Types

Type	SLA	Tunnels	Routing	Use Case
HA VPN	99.99%	2+ (required for SLA)	Dynamic (BGP)	Production
Classic VPN	99.9%	1	Static or dynamic	Legacy/simple

Setting Up HA VPN

```
# Create an HA VPN gateway
gcloud compute vpn-gateways create my-ha-vpn \
  --network=my-vpc \
  --region=us-central1

# Create a Cloud Router (required for dynamic routing)
gcloud compute routers create my-router \
  --network=my-vpc \
  --region=us-central1 \
  --asn=65001

# Create an external VPN gateway (peer)
gcloud compute external-vpn-gateways create my-peer-gateway \
  --interfaces=0=PEER_IP_1,1=PEER_IP_2

# Create VPN tunnels
gcloud compute vpn-tunnels create tunnel-0 \
  --peer-external-gateway=my-peer-gateway \
  --peer-external-gateway-interface=0 \
  --vpn-gateway=my-ha-vpn \
  --vpn-gateway-region=us-central1 \
  --ike-version=2 \
  --shared-secret=SHARED_SECRET \
  --router=my-router \
  --interface=0

gcloud compute vpn-tunnels create tunnel-1 \
  --peer-external-gateway=my-peer-gateway \
  --peer-external-gateway-interface=1 \
  --vpn-gateway=my-ha-vpn \
  --vpn-gateway-region=us-central1 \
  --ike-version=2 \
  --shared-secret=SHARED_SECRET \
  --router=my-router \
  --interface=1

# Configure BGP sessions on the Cloud Router
gcloud compute routers add-interface my-router \
  --interface-name=if-tunnel-0 \
  --vpn-tunnel=tunnel-0 \
  --ip-address=169.254.0.1 \
  --mask-length=30 \
  --region=us-central1

gcloud compute routers add-bgp-peer my-router \
  --peer-name=bgp-peer-0 \
  --interface=if-tunnel-0 \
  --peer-ip-address=169.254.0.2 \
```

```
--peer-asn=65002 \
--region=us-central1
```

VPN vs. Interconnect

Feature	Cloud VPN	Dedicated Interconnect	Partner Interconnect
Connection	Over public internet	Direct physical connection	Via service provider
Encryption	IPsec (encrypted)	Not encrypted by default	Not encrypted by default
Bandwidth	Up to 3 Gbps per tunnel	10-200 Gbps	50 Mbps - 50 Gbps
Latency	Variable	Lowest	Low
SLA	99.99% (HA VPN)	99.9-99.99%	99.9-99.99%
Cost	Lowest	Highest	Medium
Setup time	Hours	Weeks	Days
Use case	Small/medium traffic, encrypted	Large, predictable workloads	Medium bandwidth needs

6. Cloud NAT (Network Address Translation)

- Allows instances with **no external IP** to initiate **outbound** connections to the internet (e.g., downloading packages, calling external APIs)
- Does **NOT** allow inbound connections — external hosts cannot initiate a connection to a NATted VM
- Statelessly translates outbound traffic; response packets are allowed back automatically
- **Cloud Router is required** — Cloud NAT is configured on top of a Cloud Router

```
# Create Cloud Router (required by Cloud NAT)
gcloud compute routers create MY_ROUTER --network=MY_VPC --region=REGION
# Create Cloud NAT
gcloud compute routers nats create MY_NAT --router=MY_ROUTER --region=REGION --auto-allocate-nat-external-ips
# View NAT config
gcloud compute routers nats describe MY_NAT --router=MY_ROUTER --region=REGION
```

- `--auto-allocate-nat-external-ips` — GCP automatically allocates and manages external IPs for NAT
- `--nat-all-subnet-ip-ranges` — applies NAT to all subnets in the VPC; use `--nat-custom-subnet-ip-ranges` to restrict scope

7. Cloud Armor (DDoS and WAF)

- **Layer 7 security policy** attached to **external HTTP(S) load balancers** (External Application LB)
- Protects against OWASP Top 10 vulnerabilities, volumetric DDoS attacks, and allows IP allowlist/blocklist enforcement

```
gcloud compute security-policies create my-policy --description="WAF policy"
gcloud compute security-policies rules create 1000 --security-policy=my-policy --expression="not (ip.src in 192.168.0.0/24)"
gcloud compute security-policies rules create 2000 --security-policy=my-policy --action=allow
gcloud compute backend-services update MY_BACKEND --security-policy=my-policy --global
```

Key Concepts

- Rules are evaluated in **priority order** (lower number = evaluated first); first matching rule wins
- **Preconfigured WAF rules** use Google-maintained signatures, e.g.:

```
evaluatePreconfiguredWaf('xss-v33-stable')
evaluatePreconfiguredWaf('sqli-v33-stable')
```

- Security policies can have a **default rule** (lowest priority) that acts as the catch-all

Exam tip: Cloud Armor attaches to **backend services**, not to individual VMs, subnets, or firewall rules directly. It only works with external HTTP(S) load balancers, not Network LBs.

8. Custom Static Routes

- Route traffic to specific destinations via a chosen **next-hop**
- Evaluated **after** dynamic routes; within the same priority, **most specific prefix wins** (longest prefix match first); ties broken by **priority number** (lower = higher priority)

Next-Hop Types

Next-Hop	Use Case
Instance	Route through a VM (e.g., firewall appliance)
IP address	Specific internal IP as gateway
VPN tunnel	Route over a Cloud VPN tunnel
Interconnect attachment	Route over Dedicated/Partner Interconnect
Default internet gateway	Route to the public internet

```
gcloud compute routes create my-route --network=MY_VPC --destination-range=10.100.0.0/16 --priority=1000 --next-hop=10.100.0.1  
gcloud compute routes list --filter="network=MY_VPC"
```

- The **default route** (`0.0.0.0/0 → default-internet-gateway`) is auto-created in every VPC. Delete it to prevent internet access from a VPC (useful for private/restricted environments).

Exam tip: Routes are evaluated by **most specific prefix first**, then by **priority** (lower number = higher priority). A `/28` route always wins over a `/16` route for matching destinations, regardless of priority.

Exam Practice Questions

1. You need to create a VPC where you control all subnet IP ranges and don't want subnets automatically created in new regions. What should you create?
 - Answer: A **custom mode VPC**. Auto mode creates subnets automatically; custom mode gives you full control.
2. Two teams in different projects need their VMs to communicate over internal IPs. What's the simplest approach?

- Answer: Set up **VPC Network Peering** between the two project VPCs. Traffic uses internal IPs and stays on Google's network.
3. **You need to allow HTTP traffic only to instances tagged as "web-server". How should you configure the firewall rule?**
- Answer: Create an ingress firewall rule with `--allow=tcp:80 --source-ranges=0.0.0.0/0 --target-tags=web-server`.
4. **Your organization wants to enforce a common set of firewall rules across all projects. Where should these rules be applied?**
- Answer: Use **hierarchical firewall policies** at the organization or folder level. These are evaluated before VPC-level rules and are inherited by all child projects.
5. **VPC A peers with VPC B, and VPC B peers with VPC C. Can VPC A communicate with VPC C?**
- Answer: **No**. VPC peering is non-transitive. VPC A cannot reach VPC C through VPC B. You would need a direct peering between A and C.
6. **You need to connect on-premises to GCP with 99.99% SLA and encrypted traffic. Which option should you choose?**
- Answer: **HA VPN** — Provides 99.99% SLA with IPsec encryption. (Dedicated Interconnect has higher bandwidth but doesn't encrypt by default.)
7. **VMs in a subnet with no external IPs need to access Cloud Storage. What do you need to enable?**
- Answer: Enable **Private Google Access** on the subnet. This allows VMs with internal-only IPs to access Google APIs and services.
-

Glossary

ASN (Autonomous System Number) — A unique number assigned to a network or group of networks under a common routing policy, used in BGP routing to identify peer networks.

Auto Mode VPC — A VPC network mode in which Google Cloud automatically creates one subnet per region with predefined IP ranges; not recommended for production use.

BGP (Border Gateway Protocol) — A dynamic routing protocol used to exchange routing information between networks; required for HA VPN and Cloud Interconnect dynamic routing.

CIDR (Classless Inter-Domain Routing) — A method for allocating IP addresses and routing, expressed as an IP address followed by a slash and a prefix length (e.g., 10.0.0.0/24).

Classic VPN — An older GCP VPN gateway type offering 99.9% SLA that supports static or dynamic routing via a single tunnel; considered legacy compared to HA VPN.

Cloud Armor — A GCP Layer 7 security service that provides DDoS protection and Web Application Firewall (WAF) capabilities, attached to external HTTP(S) load balancers.

Cloud Interconnect — A GCP service providing high-bandwidth, low-latency direct or provider-mediated physical connections between on-premises networks and Google's network.

Cloud Logging — GCP's managed log storage and analysis service that can receive VPC Flow Log data, firewall log data, and other infrastructure logs.

Cloud NAT (Network Address Translation) — A GCP managed service that allows VM instances without external IP addresses to initiate outbound connections to the internet, while blocking unsolicited inbound connections.

Cloud Router — A GCP resource that provides dynamic routing using BGP; required for Cloud NAT and HA VPN dynamic routing.

Cloud Storage — GCP's scalable object storage service; used for storing VPC Flow Logs and as a destination for exported logs.

Cloud VPN — A GCP service that connects on-premises or other cloud networks to a GCP VPC network over an encrypted IPsec tunnel through the public internet.

Custom Mode VPC — A VPC network mode where subnets are created manually with user-defined IP ranges, providing full control over the network topology; recommended for production.

DDoS (Distributed Denial of Service) — A type of cyberattack that floods a service with traffic from multiple sources to make it unavailable; Cloud Armor helps mitigate these attacks.

Dedicated Interconnect — A GCP service providing a direct physical connection between an on-premises network and Google's network, offering 10–200 Gbps bandwidth without traversing the public internet.

Default VPC — A pre-created auto-mode VPC network present in every new GCP project, with one subnet per region and default firewall rules.

Destination Range — In firewall rules and routes, the IP address range that identifies the target destination of network traffic.

Egress — Network traffic flowing outbound from a resource (e.g., from a VM to the internet or another network).

External IP — A publicly routable IP address assigned to a GCP resource, allowing it to communicate directly with the internet.

Firewall Rule — A VPC-level policy that allows or denies specific network traffic to or from GCP instances based on protocol, port, source/destination, and target.

GCP (Google Cloud Platform) — Google's suite of cloud computing services, including compute, networking, storage, and data services.

GKE (Google Kubernetes Engine) — GCP's managed Kubernetes service; GKE pods and services require secondary IP ranges on subnets.

HA VPN (High Availability VPN) — A GCP VPN gateway type that provides a 99.99% SLA by requiring two or more tunnels and dynamic BGP routing.

Hierarchical Firewall Policies — GCP firewall policies applied at the organization or folder level that are inherited by all child resources and evaluated before VPC-level firewall rules.

Host Project — In a Shared VPC configuration, the GCP project that owns the shared VPC network and subnets.

HTTPS (Hypertext Transfer Protocol Secure) — The encrypted version of HTTP using TLS/SSL; Cloud Armor attaches to external HTTPS load balancers.

IAM (Identity and Access Management) — GCP's system for controlling who (identity) has what access (roles/permissions) to which GCP resources.

IAP (Identity-Aware Proxy) — A GCP service that provides context-aware access control for applications and VMs; allows SSH/RDP tunneling to VMs without external IPs.

ICMP (Internet Control Message Protocol) — A network protocol used for diagnostic and error-reporting purposes (e.g., ping); referenced in firewall rule configurations.

IKE (Internet Key Exchange) — A protocol used to set up security associations in IPsec; Cloud VPN tunnels support IKE version 2.

Ingress — Network traffic flowing inbound to a resource (e.g., from the internet to a VM).

Internal IP — A private IP address used within a VPC network that is not directly reachable from the public internet.

IPsec (Internet Protocol Security) — A suite of protocols for authenticating and encrypting IP traffic; used by Cloud VPN to secure tunnel traffic.

IPv4 — The fourth version of the Internet Protocol, using 32-bit addresses; GCP VPC subnets use IPv4 CIDR ranges.

Longest Prefix Match — A routing algorithm principle where the most specific IP prefix (longest subnet mask) in the routing table is selected when multiple routes match a destination.

NAT (Network Address Translation) — The process of remapping source IP addresses, allowing VMs with private IPs to communicate with external networks through a shared public IP.

Network Tag — A label applied to a Compute Engine VM instance used to apply firewall rules or routes selectively to tagged instances.

Next-Hop — In routing, the next device or gateway that a packet is forwarded to on its way to the destination; options include instances, IP addresses, VPN tunnels, and internet gateways.

OWASP Top 10 — A standard list of the ten most critical web application security risks; Cloud Armor provides preconfigured WAF rules to defend against these.

Partner Interconnect — A GCP service that provides connectivity between an on-premises network and GCP through a supported third-party service provider, offering 50 Mbps to 50 Gbps bandwidth.

Peering Connection — A network link between two VPC networks that enables resources in each VPC to communicate using internal IP addresses.

Private Google Access — A subnet-level feature that allows VM instances with only internal IP addresses to reach Google APIs and services (e.g., Cloud Storage, BigQuery).

Protocol — In networking, a set of rules governing data exchange; firewall rules specify protocols such as TCP, UDP, and ICMP.

RDP (Remote Desktop Protocol) — A Microsoft protocol for remote graphical access to Windows systems; used to connect to Windows VMs in GCP.

RFC 1918 — An Internet standard defining private IP address ranges (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16) used for internal networking.

Route — A rule that specifies how to forward packets destined for a particular IP range to a specific next-hop.

Service Account — A GCP identity associated with a VM or application (rather than a human user), used to authenticate API calls and control resource access; can also be used to target firewall rules.

Service Project — In a Shared VPC configuration, a GCP project that uses subnets from the host project's shared VPC network to create its resources.

Shared VPC — A GCP networking feature that allows a single VPC network (owned by a host project) to be shared across multiple service projects within an organization.

SLA (Service Level Agreement) — A contractual commitment from a cloud provider guaranteeing a minimum level of service availability; HA VPN offers 99.99% SLA.

Source Range — In firewall rules, the IP address range(s) from which ingress traffic originates.

SSH (Secure Shell) — A cryptographic network protocol for secure remote login and command execution on Linux/Unix systems.

Stateful Firewall — A firewall that tracks the state of network connections; GCP VPC firewall rules are stateful, so allowed ingress automatically permits the corresponding egress reply traffic.

Subnet (Subnetwork) — A regional IP address range within a VPC network; VMs and other resources are assigned IP addresses from the subnet's CIDR range.

TCP (Transmission Control Protocol) — A connection-oriented transport layer protocol; firewall rules commonly specify TCP ports such as 80 (HTTP) and 443 (HTTPS).

UDP (User Datagram Protocol) — A connectionless transport layer protocol; used in firewall rules for protocols like DNS (port 53).

VPC (Virtual Private Cloud) — A logically isolated, global network within GCP that provides networking for Compute Engine VMs, GKE clusters, and other resources.

VPC Flow Logs — A VPC feature that captures metadata about network flows (source/destination IP, port, protocol, bytes, packets) for monitoring, forensics, and security analysis.

VPC Network Peering — A GCP feature that connects two VPC networks so their resources can communicate using internal IP addresses; peering is non-transitive and requires configuration on both sides.

VPN (Virtual Private Network) — An encrypted network connection over a public network; Cloud VPN creates IPsec-encrypted tunnels between GCP VPCs and external networks.

VPN Gateway — The GCP resource that serves as the endpoint for a Cloud VPN tunnel, either HA VPN or Classic VPN.

VPN Tunnel — An encrypted logical connection between two VPN gateways over the public internet, used to securely connect networks.

WAF (Web Application Firewall) — A security layer that filters and monitors HTTP traffic to protect web applications from attacks; Cloud Armor provides WAF capabilities.